Show that:

- $r_{i1} = -[\delta L_i / \delta \hat{y}_i] = -\frac{\delta(log(1+e^{\hat{y}_i}) - y_i \hat{y}_i)}{\delta \hat{y}_i}$
- Is equal to $-[p_i - y_i]$

Recall that the derivative is (from the previous slides):

- $r_{i1} = -[\frac{e^{\hat{y}_i}}{1+e^{\hat{y}_i}} - y_i]$

Recall that

- $\hat{y}_i = \log \frac{p_i}{1-p_i}$
- So $\rightarrow e^{\hat{y}_i} = \frac{p_i}{1-p_i}$
- $\rightarrow e^{\hat{y}_i} - e^{\hat{y}_i} p_i = p_i$
- $\rightarrow e^{\hat{y}_i} = p_i + e^{\hat{y}_i} p_i$
- $\rightarrow e^{\hat{y}_i} = (1 + e^{\hat{y}_i}) p_i$
- $\rightarrow \frac{e^{\hat{y}_i}}{1+e^{\hat{y}_i}} = p_i$

Thus:

- $r_{i1} = -[\frac{e^{\hat{y}_i}}{1+e^{\hat{y}_i}} - y_i] = -[p_i - y_i]$